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3D Look Ahead Electromagnetic Inversion in Near Vertical Wells – A Bridge Between Vertical and Horizontal Wells 3D Geology Understanding

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Abstract

1D look ahead inversion for ultra deep azimuthal resistivity (UDAR) tools has been demonstrated in vertical and near vertical wells, allowing detection of resistivity ahead before a boundary is intersected, optimizing the setting of casing or coring points. Horizontal look ahead UDAR inversion is frequently requested; however, this requires 3D inversion with look ahead sensitivity to account for variability in resistivity encountered. A bridge between these technologies is look ahead 3D inversion in a near vertical well.

UDAR tools acquire data while rotating and measure the full 3D electromagnetic (EM) field while drilling. Models generated match predicted data to measured data to generate a representation of the subsurface. Inversions often simplify modelling to limit the number of unknowns, the most common approach treating the subsurface as a layer cake scenario with 1D inversion. The advent of 3D inversion enables mapping fully 3D geological scenarios. This approach, combined with a tool with look ahead sensitivity in near vertical wells where geology is simple, is demonstrated and the results verified.

The detection of a high resistivity salt prior to entry and detection of low resistivity inclusions within the salt was the ideal environment to test a new look ahead technology. This represents both ends of the resistivity spectrum with regards to applications for the setting of casing points prior to breaching a boundary. Initial tests were for a 1D inversion which has since been verified in other fields, and was identified as an ideal data set to test the new 3D look ahead inversion.

The salt entry is a simple boundary from low to high resistivity with the vertical well perpendicular to the target boundary. 3D inversion results using the full resistivity tensor show the boundary 45ft below the UDAR transmitter 36ft ahead of the bit. The 3D inversion additionally identified low resistivity inclusions approximately 70ft ahead of the transmitter, with increased depth of detection due to the surrounding high resistivity environment. Due to simplification of the modelling the 1D look ahead inversion identifies these boundaries earlier, however the 3D inversion showed that the inclusions are not simple horizontal beds but have a more complex morphology. Entry to all boundaries was confirmed as the well intersected the target units with shallow logging while drilling (LWD) measurements.

Full 3D inversion is the key to look ahead inversion in horizontal wells, demonstrating this ability in a near vertical well, is key to taking the technology beyond horizontal wells. 3D inversions can allow Geoscientists

to have a mechanism to validate the assumptions used while building the conceptual geological model with sparse data and get better understanding of rock and fluids properties distribution.

Introduction

1D Look Ahead Inversion

In near vertical wells one dimensional (1D) look ahead inversions are deployed to identify resistivity boundaries ahead of an ultra-deep azimuthal resistivity (UDAR) transmitter (Ma et al. 2022). In an optimal scenario, the transmitter is positioned approximately 9ft behind the bit so that resistivity boundaries ahead of the bit can be identified allowing for geostopping prior to their intersection (Sinha et al 2023). Resistivity changes ahead of the bit can indicate a change in formation or fluids, which signals potential changes in pore pressure, rock stability or fluid composition (Eaton 1975, Qatari 2024). Intersecting these boundaries can pose a hazard to drilling operations, for example the potential to cause a kick if a zone is over pressured or for severe mud losses in under pressure zones.

Electromagnetic (EM) energy transmitted by the UDAR tool extends both ahead, around and behind the tool, the measurement and subsequent inversion of the EM field allows formation and fluids surrounding and ahead of a tool to be investigated (Fig. 1).

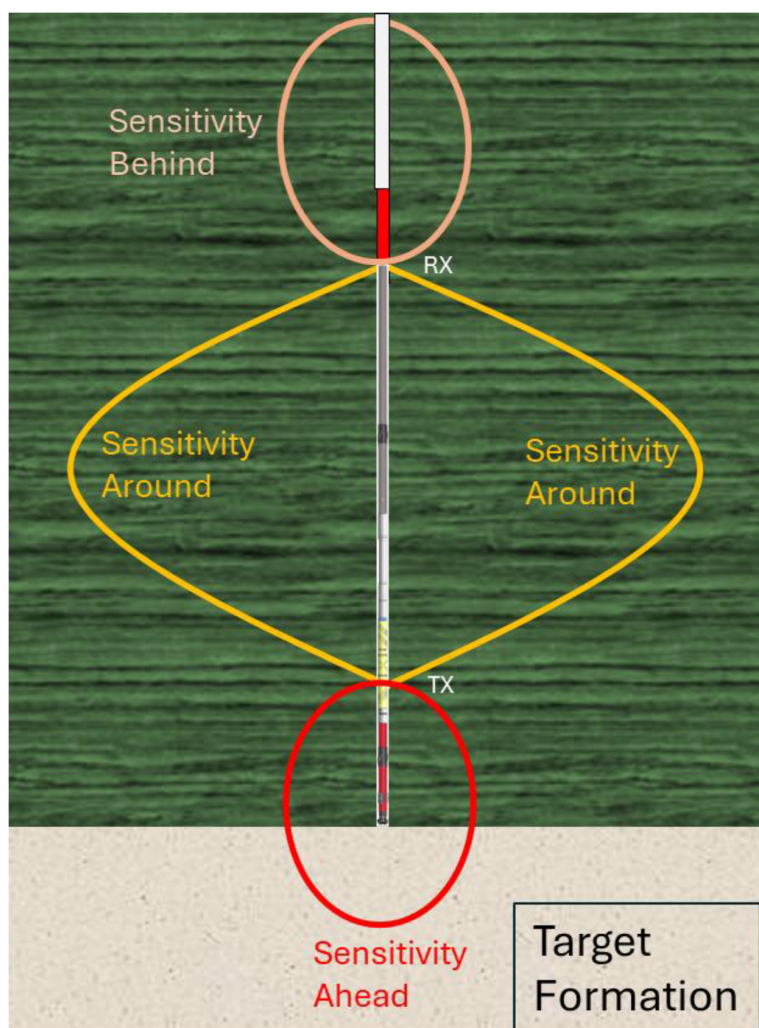


Figure 1—Depiction of electromagnetic energy projected around a UDAR tool.

The EM signal from ahead of the tool is weaker than that projected around the tool. To enhance and identify its contribution to the EM field 1D lookahead inversion accounts for the contribution of the EM field from the formation and fluids surrounding and behind the tool, by creating a model for the formations and fluids that have already been penetrated based on shallow tool readings (Ma et al 2022). The expected contribution of the resultant model to the EM field can be generated and extracted from the measured data. The remaining measurement represents the contribution of the formation and fluids ahead of the tool, a 1D inversion algorithm can then represent this as resistivity layers ahead of the transmitter.

As a boundary ahead is approached, initially the contribution of the change in resistivity ahead to the electromagnetic (EM) field is small, and can result in a response with high uncertainty, both in the position of the boundary and the resistivity across the boundary. With increased proximity to the boundary the contribution of the bounding formation increases, and uncertainty decreases to provide both the position of the boundary and the resistivity beyond it. Where the overburden has low resistivity the distance that a boundary ahead can be detected is decreased, related to decreased propagation of the electromagnetic field. In a high resistivity section, the depth of detection is increased as the EM energy is propagated further. In all scenarios it is therefore critical to understand what resistivity environment that one of these tools will be deployed in and the potential range of resistivities that will be encountered.

As with inversion in horizontal wells any resistivity heterogeneity can negate, the 1D assumption, that resistivity changes only happen in the vertical plane, this will increase uncertainty in the result (Wu et al 2018, Clegg et al 2018). In look ahead wells the assumption that drives the modelling of the formations surrounding and behind the transmitter is that the formation boundaries are flat and that there is no lateral heterogeneity in the formation resistivity. A small amount of formation dip (up to 10 degrees) can be tolerated, likewise well inclinations up to 30 degrees produce limited distortion of the inversion results. Increased well inclination or formation dip causes increased uncertainty in the inversion results and the error on the position and resistivity of the target boundary increases. Any lateral heterogeneity in the formation resistivity will result in additional uncertainty, such as major lateral changes driven by geological variation in the section above the target driven by the depositional environment or post deposition deformation. Under these circumstances a 1D inversion is not sufficient to enable accurate identification of a resistivity contrast ahead. Where it is critical to identify a boundary before it is penetrated as it poses a hazard to drilling this uncertainty becomes a significant risk.

3D Inversion

Three dimensional inversions have been deployed in horizontal wells to identify lateral heterogeneity in resistivity boundaries allowing mapping of geobodies to the side of the well (Wilson et al 2018, Clegg et al 2018, Ameneiro et al 2023). These inversions have been deployed real time to identify lateral changes in resistivity and steer the well azimuthally, to optimise the well path within the stratigraphy (Elkhamry et al 2023). This suggests that there is the potential to establish the resistivity surrounding the well bore in a similar way to establishing the EM field generated in formations surrounding and behind the transmitter in 1D look ahead inversions and then subtract this contribution from the whole EM field to isolate the contribution to the EM field from the formations ahead.

Examination of the raw component data from horizontal wells that showed major resistivity changes ahead in a scenario where casing was present ahead of the transmitter demonstrates that major changes ahead have a major impact on the EM field in the region of 50ft before the resistivity change is passed (Clegg et al 2021). In a 3D EM field isolating the contribution of the field surrounding and behind the tool for a true 3D inversion becomes difficult. As the inversion is already inverting a volume along the axis of the BHA. Inverting the full volume including the contribution from ahead of the tool is simpler and produces a better result.

A 3D look ahead inversion has been developed for horizontal wells, to provide earlier warning of resistivity changes during geosteering operations (Elkhamry et al 2025, Ma et al 2025). As this is a true

3D inversion the orientation of the well path can be at any inclination, and therefore the technology is now applied in a vertical well to explore its use for early warning in scenarios where the 1D look ahead inversion may be degraded by failure in the 1D assumption.

3D Look Ahead Inversion

Inversion-based resistivity imaging often suffers from ambiguities due to the underdetermined nature of the problem: the number of unknowns in the model space significantly exceeds the available independent measurements in the data space. This challenge is particularly pronounced in 3D inversions, where the subsurface is discretized into hundreds of thousands of unknown resistivity values, yet the measurement data remains limited. Furthermore, measurement noise and calibration errors compound the difficulty, reducing inversion reliability.

The primary requirement for enabling effective 3D look-ahead inversion is ensuring adequate signal sensitivity to resistivity variations in all three spatial dimensions, especially around and ahead of the wellbore. To address this, two critical innovations have been implemented to enhance signal quality and sensitivity:

1. **Multi-collar UDAR Tool Design:** The development of advanced ultra-deep azimuthal resistivity (UDAR) tools has significantly improved measurement diversity and sensitivity (Constable et al., 2012; Wu et al., 2018). These tools employ a tilted transmitter mounted on the transmitter collar and three receiver antennas spaced azimuthally at 120°, capturing full magnetic tensor components while rotating during drilling. Operating across a frequency range of 2 kHz to 64 kHz with spacing between 20 ft and 133 ft, this design enables comprehensive azimuthal and axial sensitivity—maximizing the information content from magnetic dipole sensors.
2. **In-situ Calibration in Vertical/Low-angle Wells:** In relatively homogeneous or layered formations, shallow measurements located just behind the transmitter can be extrapolated laterally to establish a reference formation profile. This known profile serves as a basis for in-situ calibration of the full magnetic tensor components before approaching the target zone, reducing systematic errors and enhancing the inversion's ability to resolve weak signals from ahead of the bit.

To support real-time 3D inversion under these constraints, a dedicated inversion engine was developed using a Gauss-Newton optimization algorithm in combination with octree-mesh discretization (Wilson et al., 2018). The Gauss-Newton method is well-suited for nonlinear inverse problems, offering quadratic convergence and efficient reduction of data misfit through iterative updates. Octree meshing adaptively refines the model grid around critical regions (e.g., near the borehole) while coarsening it elsewhere, thereby reducing computational demands without sacrificing resolution (Yang et al., 2013; Haber and Schwarzbach, 2014). The forward modeling component of the inversion process leverages the finite volume method (FVM), which conserves flux across cell boundaries, ensuring numerical accuracy even in irregular geological formations. This is particularly important for capturing the intricate details of faults, fractures, and other complex subsurface structures.

The 3D inversion optimizes the below cost functions with data misfit and model misfit:

$$\vec{m}^* = \text{minimize}(\phi_d(\vec{m}) + \beta\phi_m(\vec{m})) \quad (1)$$

Where ϕ_d quantifies the discrepancy between predicted and observed data, ϕ_m represents the model objective function, incorporating L2-norm regularization to enhance noise robustness and stabilize convergence. The regularization includes a zeroth-order term to gauge inversion initially from a single resistivity value from shallow measurement right behind the deep transmitter, and a first-order term to ensure smoothness and model-based continuity. This approach avoids imposing predefined geological structures, ensuring that any resolved 3D features are entirely data-driven and unbiased.

1D vs. 3D Inversion Results

1D Look Ahead Inversion Results. A UDAR look ahead configured tool was deployed with a 40ft spacing from the transmitter to receiver 1 and 80ft spacing to receiver 2. Low frequencies of 2 and 4 kHz were used for the 1D look ahead inversion for early detection of the top of a salt formation (Fig 2).

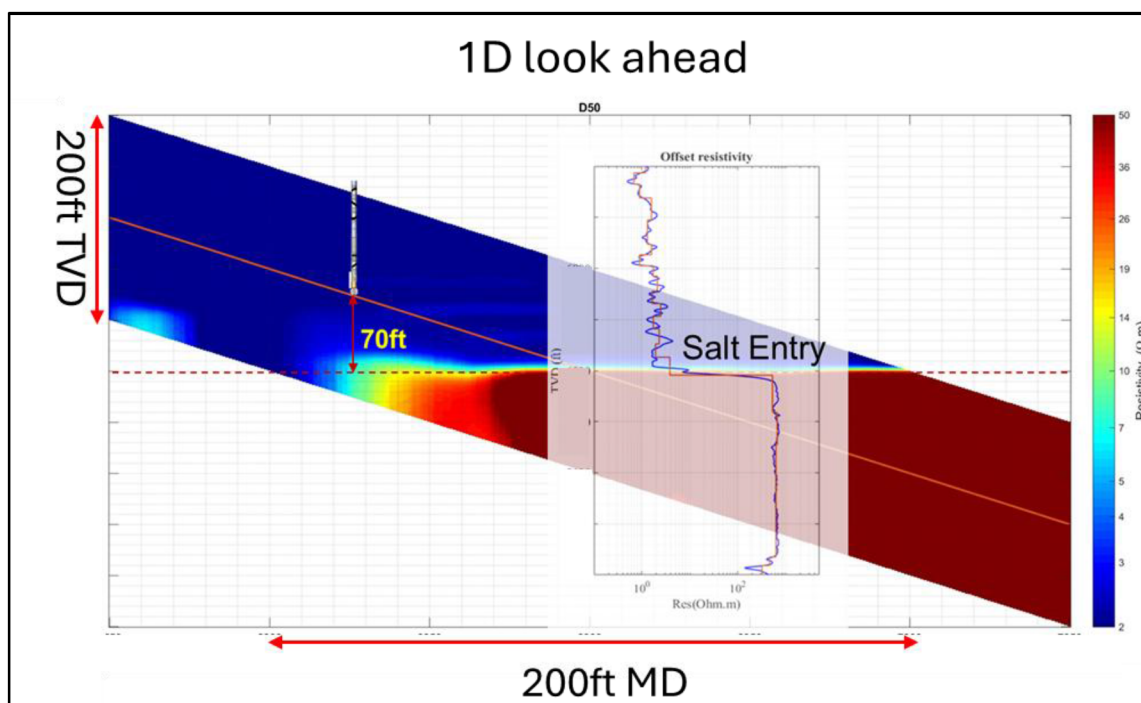


Figure 2—One dimensional look ahead inversion. The well is vertical. RX1 positioned 40ft behind the transmitter, RX2 positioned 80ft behind the transmitter. 2 and 4 kHz frequencies were used for both receivers.

First detection of the top reservoir is 70ft TVD below the transmitter with approximately 10ft uncertainty on the position of the boundary and high uncertainty in the resistivity of the formation beyond the boundary. This is typical of early detection in 1D inversions seen in both horizontal and vertical well examples (Ma et al 2022). This represents the edge of the envelope for the EM field extending below the tool. As the tool gets closer to the boundary and an increased volume of the formation below comes within the envelope, the uncertainty on the boundary position and resistivity decreases. Best practice during look ahead operations is to establish the position of a boundary over a minimum of 4 readings to confirm any uncertainty and accurately establish the position of the boundary. In this scenario with first detection of the boundary 70ft below the transmitter there is plenty of distance to clearly establish the boundary position. As this detection is distance from the transmitter, its position in the BHA has a major impact on mapping the boundary before it is penetrated. The latest generation of UDAR tools now allow the transmitter to be placed only 9ft behind the bit (Ma et al 2022) in both horizontal as well as vertical wells (Sinha et al 2023).

Multiple variables have an impact on the depth of detection, the most significant being: Transmitter to receiver spacing, frequency of transmission, host formation resistivity, target unit resistivity and the sharpness of the boundary. Bigger transmitter to receiver spacing will increase depth of detection up to a limit where noise can degrade the measurement. Lower transmission frequencies will similarly increase depth of detection. Low host formation resistivity will decrease depth of detection while high host resistivity will increase it. Additionally, the sharper the resistivity contrast across the target boundary the earlier it will be detected with increased certainty. The impact of these variables means that simple depths of detection cannot be quoted with accuracy and that modelling needs to be conducted to assess the influence of all these factors.

3D Inversion Results

The entry into the salt represents an ideal scenario to deploy a 1D inversion. The well path is near vertical, and the top boundary has a high contrast flat boundary. The overburden section shows low resistivities which will decrease depth of detection, but the boundary has a sharp very high resistivity contrast. The boundary was intersected during this test section and the accuracy of the boundary pick was excellent.

With the development of the 3D look ahead inversion, originally developed for horizontal wells, this data set was tested to investigate the accuracy of the 3D look ahead results. The position of the boundary is known as it has been penetrated, and the 1D look ahead inversion results demonstrate that the UDAR tool is clearly sensitive to the boundary ahead, so this is the ideal scenario to also test the 3D look ahead inversion in a controlled environment.

Results from the 3D look ahead inversion show the first detection of the salt 45ft ahead of the transmitter (Fig 3).

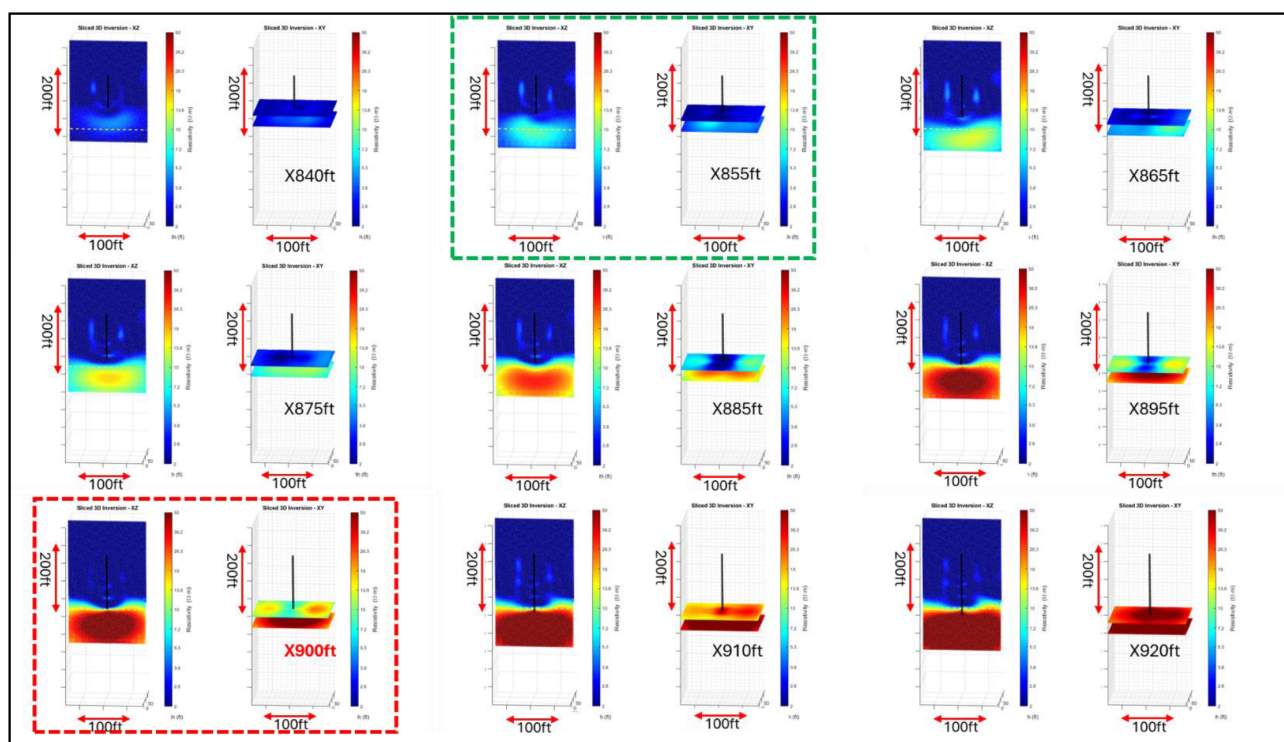


Figure 3—3D look ahead inversion results as the top salt boundary is approached and then crossed.

In each image the left panel shows a view sliced along the well path, the right hand image shows planes perpendicular to the well path. The depths represent the transmitter position at each measure point. The green box indicates the first good detection of the top, the red box the interception point.

The boundary detection for the 3D inversion at 45ft TVD is later than the 70ft TVD for the 1D inversion. This is because this is a 1D scenario and the 1D assumptions work perfectly in this scenario, therefore the enhancements for the 1D inversion to isolate the signal ahead of the transmitter improve the result here when compared to the 3D result which is inverting the whole volume. However, the 3D inversion is successful and detects the boundary at the same depth as the 1D inversion look ahead inversion and its position was confirmed when it was penetrated. In this scenario the 1D inversion would always be expected to produce earlier detection, the only reason to deploy the 3D inversion would be if lateral heterogeneity is expected.

Signal sensitivity shows that multiple components show significant variability upon approach to the boundary, as a precursor to its interception (Fig. 4). This represents the contribution to the EM field from the geology ahead of the transmitter. Where there is no change in resistivity ahead the component data would not show this deflection in the data.

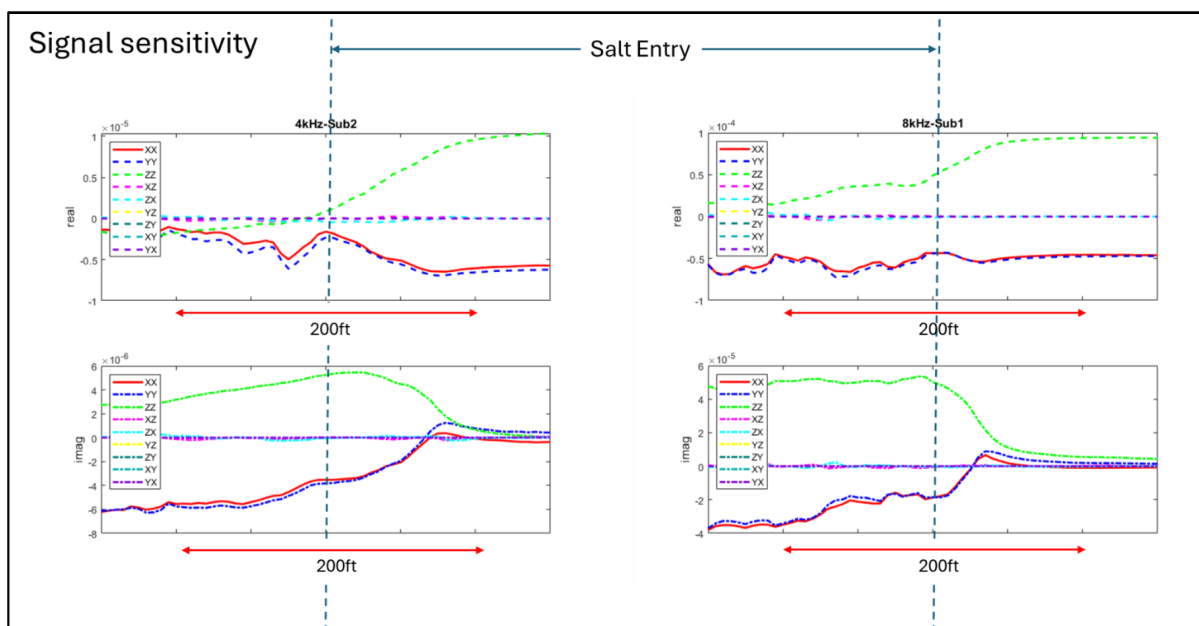


Figure 4—Signal sensitivity across the salt boundary.

Salt Inclusions

Within the salt are carbonate inclusions, tool measurements indicate that these are not 1D and result in increased uncertainty on the 1D inversion result (Fig. 5). Only when the tool gets very close to them does the uncertainty of the boundary position decrease. The 3D nature of these inclusions was not anticipated but provides an opportunity to map their 3D distribution with the 3D look ahead inversion. An initial 3D look ahead inversion using only the same 5 components as used for the 1D inversion shows that there appears to be lateral heterogeneity in these geobodies (Fig.5). The 5 components only 3D inversion shows some instability but clearly indicated why the 1D inversion is struggling to pick the boundaries correctly.

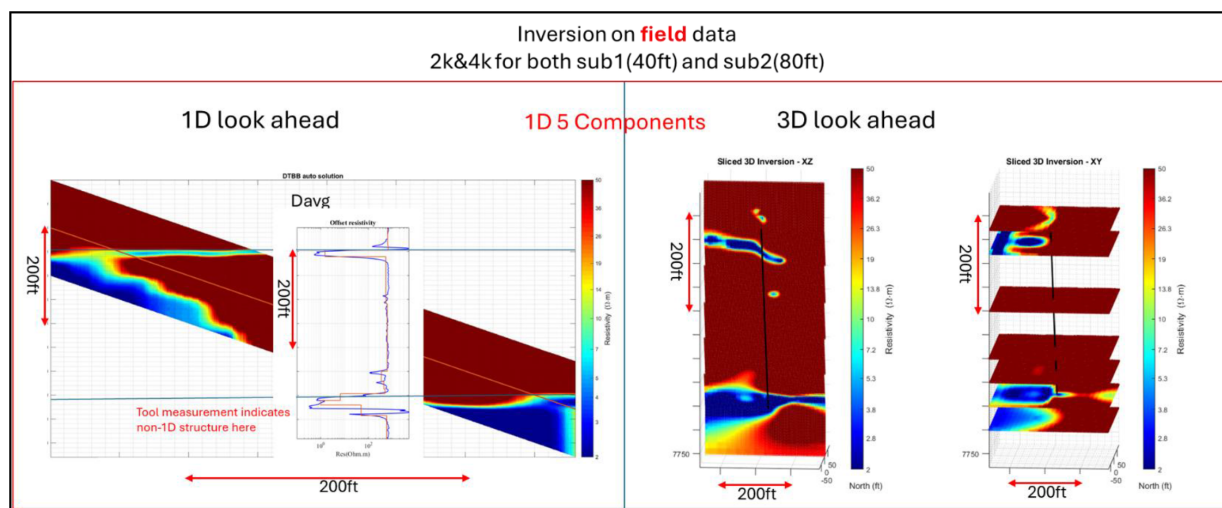


Figure 5—Left side picture, 1D look ahead inversion result for the carbonate inclusions. Right Side picture, 3D look ahead inversion using only 5 components.

A full 9 component 3D inversion of the interval shows the complex distribution of these geobodies (Fig. 6). Azimuthal resistivity images that are independent from the components used in the 3D inversion also show that these geobodies have a complex distribution and why the 1D inversion is failing. The azimuthal resistivity images are a direct measurement of resistivity surrounding the well bore. If boundaries are

encountered directly below these will show symmetrical images starting in the centre of the image and spreading to the sides. Both inclusions show lateral heterogeneity.

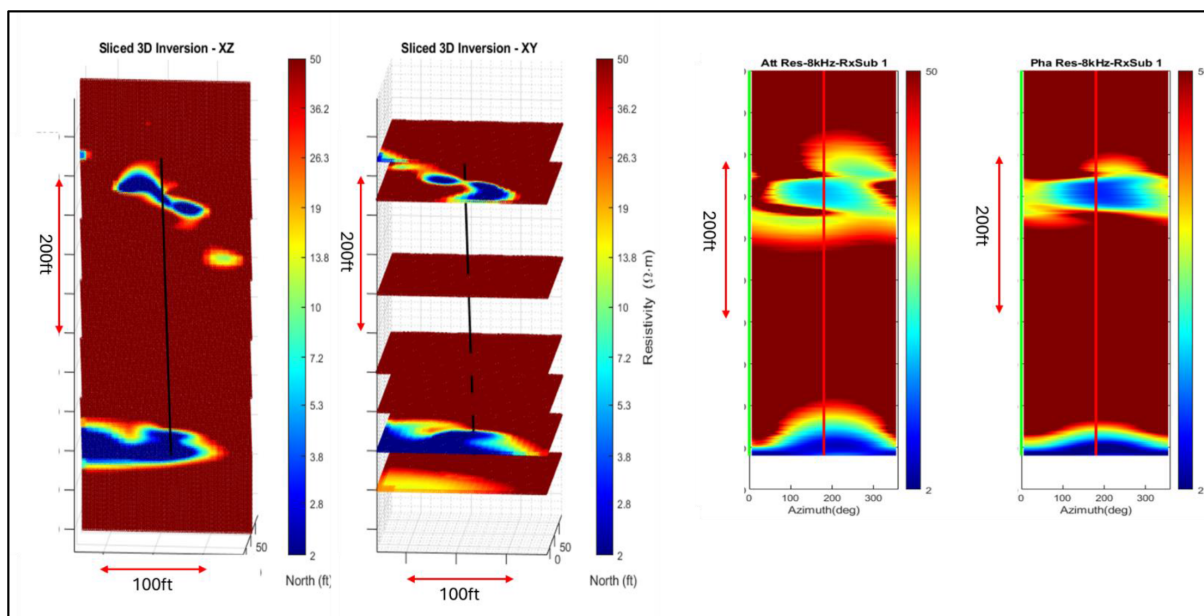


Figure 6—Left side picture, 3D look ahead inversion result for the carbonate inclusions All 9 Components, View sliced along the well path and planes perpendicular to the well path. Right Side pictures, azimuthal resistivity images.

The first of the two inclusions is smaller, and the tool and inversion will be less sensitive to it and therefore have limited sensitivity ahead, however the 2nd inclusion is much larger and the sensitivity to is ahead can be shown (Fig 7). This 2nd inclusion can be first detected 70ft ahead of the transmitter (Fig. 7).

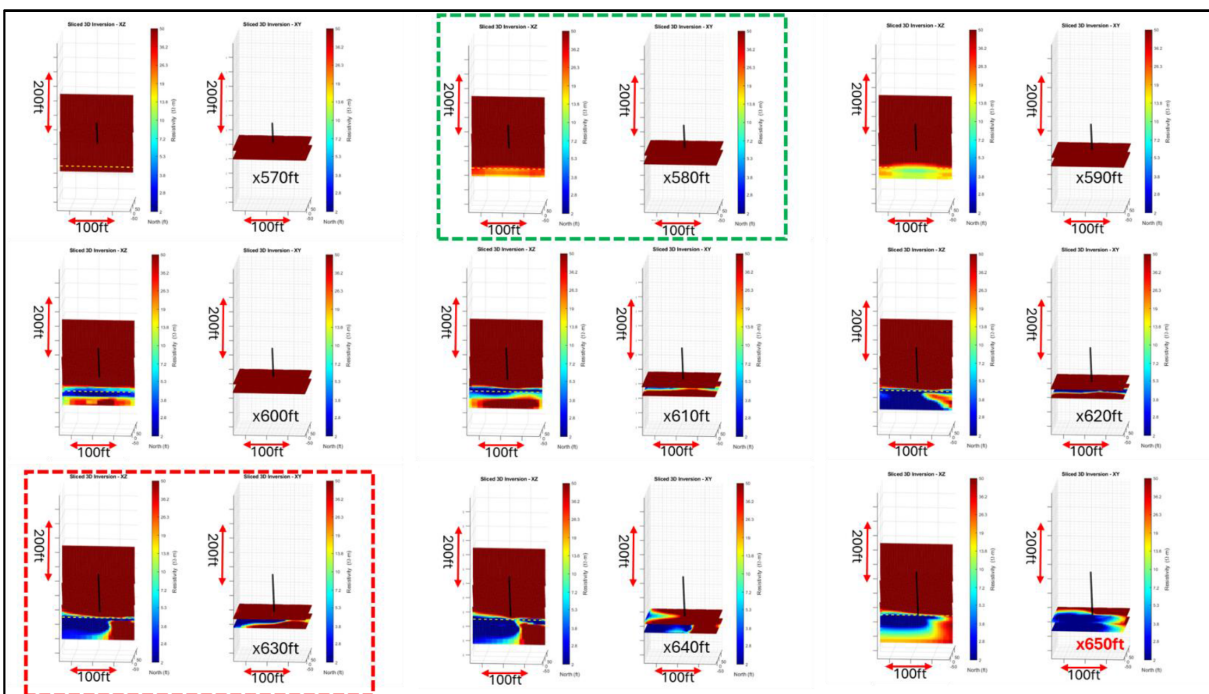


Figure 7—3D look ahead inversion results as the 2nd carbonate inclusion is approached. The depths represent the transmitter position at each measure point.

The increase in look ahead sensitivity to the carbonate inclusion to 70ft, from the 45ft detection of the top salt is because the formation resistivity is much higher and therefore the EM energy is transmitted further.

Conclusions

One dimensional look ahead inversions show more sensitivity ahead when the 1D assumption that there is no lateral heterogeneity is true. In many wells this is a valid assumption and deploying 1D look ahead EM technologies allows boundaries ahead of the UDAR transmitter to be detected before they are penetrated. Early warning of pressure, formation and fluid changes ahead based on observed resistivity changes allows drilling to be stopped prior to boundaries being intersected.

In complex geological scenarios where there is lateral heterogeneity in resistivity the 1D assumption is no longer valid and the 1D look ahead inversion results can be unreliable. In these scenarios as demonstrated here a full 3D look ahead inversion is required to reliably detect resistivity changes ahead. Simultaneously mapping the 3D distribution of resistivity surrounding and ahead of a tool allows hazards to be identified, complex geobodies to be mapped and geostopping prior to the intersection of resistivity boundaries.

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